

Time to Commit: Facilitating Open Science Practices Using GitHub

Oliver Collins



About Me



Studying **BSc Psychology with Clinical Approaches** at Sussex

- Recently completed my Placement Year as a Research Assistant at the **Reality Bending Lab**, investigating the neuropsychology of reality perception
- Current **Junior Research Associate** in the School of Psychology
- **Joint-winner**: 2025 Jane Beattie Prize for the Best Overall 2nd Year Performance
- **Winner**: 2024 Lisa Rodrigues Prize for the Best Overall 1st Year Performance

Academic interests spanning perception, psychophysiology, research methods, and programming

- Developed four psychology-based Python packages: *PyChroma* (dot-based chromostereopsis illusion generator), *PyGrid* (grid illusion generator), *AfterPy* (afterimage illusion generator), and *Resp-Music* (respiration-controlled music interface)



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Psychological Research as a Black Box

Two issues across psychological research



Is psychology suffering from a replication crisis?
What does "failure to replicate" really mean?

Scott E Maxwell¹, Michael Y Lau², George S Howard¹

Affiliations + expand

PMID: 26348332 DOI: [10.1037/a0039400](https://doi.org/10.1037/a0039400)

✗ Implications for the **replicability** and **reproducibility** of findings

GitHub as a practical solution!

Introducing GitHub



What is GitHub?

- Online platform for users to upload and share their code and data files
- Tool utilised by software developers
- Lots of scope for collaboration

Demystifying GitHub

- ✗ GitHub is only for programmers
- ✗ GitHub is only for collaborative projects
- ✗ GitHub requires a strong prior technical proficiency

Basic Features

- **Repositories**: folder for a project
- **Branches**: different versions of a repository
- **Pull requests**: proposed changes made by users
- **Commits**: tracking of each saved change to a repository

MYTH VS REALITY

Introducing GitHub (Cont.)



RealityBending

Q Type / to search

Overview Repositories 56 Projects 1 Packages Teams 3 People 13

Reality Bending Lab (ReBeL)

We study the neuropsychology of reality and its distortions

Unfollow

45 followers <http://realitybending.github.io/> @RealityBending_ D.Makowski@sussex.ac.uk

README .md

Reality Bending Lab

Reality Bending Lab (ReBeL)

Where we forge the tools to help us understand the world

Welcome to our GitHub 🙌

You can find here all the open-source projects that the lab is developing. Check out our website to know more about who we are, what research we do, and access our publications:

<https://realitybending.github.io/>

Useful Reads

View as: Public

You are viewing the README and pinned repositories as a public user.

People

Top languages

HTML JavaScript Python R

TeX

Most used topics

Manage

neuroscience personality psychology

ecg illusion



Introducing GitHub (Cont.)



Using GitHub as a researcher...

Benefits

- ✓ Storage space!
- ✓ *Efficient* collaboration
- ✓ Version control
- ✓ Hosting experiments
- ✓ **Open science...**

Caveats

- ✗ Intimidating
- ✗ Jargony
- ✗ Documentation not always the most helpful



GitHub at the Reality Bending Lab



A Typical Repository



RealityBending / IllusionGamePhenomenologicalControl

Code Issues 1 Pull requests Agents Actions Projects Security and quality Insights Settings

main 2 Branches

AnafNeves added tracked changes doc 2 days ago

- analysis
- data
- documents
- experiment
- paper
- .gitignore
- README.md
- index.html

Procedure

The experiment's setup follows the born-open principle (De Leeuw, 2023), and all its related material is available at <https://github.com/RealityBending/IllusionGamePhenomenologicalControl>. The online experiment, implemented entirely using *JsPsych* (De Leeuw, 2015), has its code stored on GitHub and leverages the power of the platform to host the experiment for free. Participant's raw

Releases

No releases published
[Create a new release](#)

Deployments 114

github-pages 2 days ago
[+ 113 deployments](#)

Packages

No packages published
[Publish your first package](#)

Illusion Sensitivity & Phenomenological Control

Abstract. Visual illusions highlight how easily our conscious experience can be altered with respect to perceptual

GitHub and Open Materials



  main  IllusionGamePhenomenologicalControl / experiment / 

 OliverACollins Update demographics.js 

Name	Last commit
..	
images	fixed is
stimuli	added
IllusionGame.js	fixed is
demographics.js	Update
index.html	Fixes fo
instructions.js	fixed is
questionnaires.js	change

README

Illusion Sensitivity & Phenomenological Control

Abstract. Visual illusions highlight how easily our conscious experience can be altered with respect to perceptual reality. Despite sharing in-principle mechanisms with phenomenological control, i.e., the ability to alter our perceptual experience to match task demands or expectations, research tying the two remains scarce. This study aimed to replicate and extend Lush et al., (2022) by testing for a relationship between phenomenological control and illusion sensitivity across multiple illusion types, assessing the role of higher-level (top-down) processes. It also examines associations between visual illusion sensitivity and psychoticism to assess the potential contribution of lower-level effects (e.g., weak priors). Five hundred participants were recruited in an online study using an illusion-based perceptual decision paradigm (Illusion Game; Makowski et al., 2023) measuring sensitivity to three visual illusions (Ebbinghaus, Müller-Lyer, Vertical-Horizontal), alongside the Phenomenological Control Scale and personality measures. Bayesian correlation analyses provided evidence supporting the absence of a relationship between phenomenological control and illusion sensitivity, replicating prior findings. No evidence was also found for an association between illusion sensitivity and psychoticism. These findings support the hypothesis that visual illusions are to some degree cognitively impenetrable, and suggest that illusion susceptibility operates independently of higher-level control over subjective experience. However, low consistency across illusion measures highlights important methodological limitations in capturing individual differences related to illusion sensitivity.

- [Try the experiment](#)
- [Registered Report Preprint](#)
- [Data](#)
- [Stage 2 Manuscript](#)
- [Data Cleaning Report](#)
- [Data Analysis Report](#)



Concluding Remarks



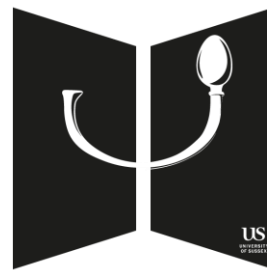
GitHub can help to...

- Enhance **transparency** across research
- Improve the **replicability** and **reproducibility** of findings
- Make you feel **empowered** as a researcher
- Strengthen your **online presence** and **academic prowess** 💪

If you are a researcher working with code/software/easily-shareable data, there is NO reason you should not use GitHub!

Thank you for listening!

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Reality
Bending
Lab